

Purujit Kantiya

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Summary

Third year Mechatronics Engineering student with 2.5 years of experience working in design teams and professional environments. My roles on the Aerospace design team required me to lead students in the design and validation of drone subsystems. This involved the use of KiCAD, OnShape, Python and custom test setups for design and validation. My current role as a verification intern has exposed me to professional testing practices including checking against requirements derived from an ICD, automation and result documentation.

Education

B.A.S.c - Mechatronics and Robotics Engineering Sep 2024 – May 2028
Grade: 4.0 Queen's University

- Awards: Deans List, Ontario Engineering Competition 1st Place, Canadian Engineering Competition 3rd Place
- Extracurriculars: Aerospace Design Team, Hyperloop Design Team, QHacks, Merlin Neurotech

Professional Experience

Verification Engineering Intern- Xona Space Systems May 2026 – Aug 2026

- Developing Python-based test scripts to verify user equipment compatibility with Xona's PULSAR signal, ensuring compliance with ICD specifications.
- Automating hardware test plans to streamline the verification pipeline and improve testing efficiency.
- Migrated hardware verification requirements from Excel to Flow Engineering and developed Python API integrations to synchronize requirements automatically.

Construction Engineering Intern - Tatham Engineering May 2025 – Aug 2025

- Reviewed contractor submittals including construction schedules, shop drawings, and material specifications, to ensure compliance with project requirements.
- Collected and analyzed survey data to confirm accuracy of contractor work and support progress verification.
- Provided technical guidance and recommendations to contractors on construction methods, sequencing, and compliance with municipal standards.

Extracurricular Experience

Captain - Queen's Aerospace Design Team May 2026 – Present

- Leading 100 students in the development of several aerospace projects ranging from research papers to physical drones for submission to international competitions and conferences..
- Responsible for maintaining team resources including money, material and faculty expertise so that all technical goals can be met.

Simulation Team Member - Queen's Aerospace Design Team Sep 2025 – June 2026

- Contributed to the development of a multi-drone swarm simulation for presentation at the 2026 ICUAS conference, using ROS2, Gazebo, and a Linux-based workflow.
- Designed and implemented ROS2 nodes that command the drones to scan and extract data from the surrounding environment.
- Used Git for version control, collaborative development, and code review to ensure reliable and maintainable simulation software.

Systems Integration Team Manager - Queen's Aerospace Design Team May 2025 – April 2026

- Leading a team of 10 students in the design, integration, and testing of electrical systems for a fixed wing UAV.
- Plan and conduct systems tests using custom test stands, sensor packages and software.
- Oversee component selection, procurement, and cost tracking to stay within project budget and timeline.
- Coordinate with structures and aerodynamics teams to align system requirements and optimize aircraft performance.

Technical Skills

Design: Solidworks CAD, Onshape, LTSpice, Altium, QGroundControl, KiCad, Altera Quartus II, Arduino

Programming: Python, C++, VHDL, ROS2, Linux, Docker, Git, Assembly, MATLAB

Productivity: Word, PowerPoint, Excel

Languages: Sindhi (Fluent), English (Fluent)